

Juan David Muñoz-Bolaños

R&D Optical Engineer | Extreme precision restoration

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photo.jpg

SUMMARY

Optical physicist specializing in high-precision instrumentation, with deep expertise in **Adaptive Optics**, **Two-Photon Microscopy**, and **Femtosecond Interferometry**. Proven track record of building complex optical setups from scratch, automating experimental control, and performing custom wave-propagation simulations. Passionate about pushing the limits of imaging resolution and precision.

Personal Interests: Outside the lab, I find joy in a balanced life through cycling, skiing, and social dancing.

EDUCATION

Medical University of Innsbruck

Ph.D. Advanced Microscopy

Innsbruck, Austria

Jan 2022 – Present

- **Thesis:** Deep tissue imaging by wavefront shaping.
- **Key Tech:** optical interferometry, spatial light modulators, optical resolution restoration strategies.
- **Achievement:** Development of phase retrieval framework to measure aberrations deep in scattering media using two-photon microscopy.

Friedrich Schiller University Jena - Abbe School of Photonics

M.Sc. in Photonics

Jena, Germany

Oct 2019 – Dec 2021

- **Thesis:** Design of a 1064 nm fiber probe Raman imaging spectrometer.
- **Focus:** Optical Design, Fiber Optics, Spectroscopy.

University of Cauca

B.Sc. in Physics Engineering

Popayan, Colombia

Mar 2012 – Mar 2018

EXPERIENCE

École Polytechnique

Research Intern (LOB)

Paris, France

Jul 2025 – Aug 2025

- Implemented variable Numerical Aperture (NA) control to enhance signal-to-noise ratio in nonlinear imaging.
- Optimized detection paths for scattering environments.

Leibniz Institute of Photonic Technology (IPHT)

Assistant Researcher

Jena, Germany

Jun 2020 – Dec 2021

- Conducted vibrational imaging (Raman/IR) for cancer tissue diagnostics.
- Integrated photothermal microscopy setups for microplastic detection.

INTECOL S.A.S.

Machine Vision Engineer

Medellin, Colombia

Mar 2018 – Mar 2019

- Designed industrial quality control systems using high-speed cameras and custom optics.
- Developed image processing algorithms for automated defect detection (Production Line).

TECHNICAL SKILLS

Optical Engineering

Design/Alignment of Multiphoton Microscopes, Adaptive Optics (SLM/Deformable Mirrors), Interferometry (Mach-Zehnder/Michelson), Ultrafast Lasers, Spectrometers.

Simulation & Analysis

LabVIEW (FPGA), Python, Optical Wave Propagation Simulations.

Languages

English Proficient, German speakable, Spanish (Native).

SELECTED PUBLICATIONS & PRESENTATIONS

- **J. D. Muñoz-Bolaños**, et al., "Scatter compensation for enabling deep two-photon microscopy using a MEMS phase light modulator," *Proc. SPIE* (2026).
- **J. D. Muñoz-Bolaños**, et al. "Wavefront corrections over large fields of view via beam cone tomography." *Proc. SPIE*, (2025).
- **J.D. Muñoz-Bolaños**, et al. "Confocal Raman Microscopy with Adaptive Optics." *ACS Photonics*, (2024).

SCHOOLS & AWARDS

ZEISS Summer School (Oberkochen, Germany)	2025
SPIE Biophotonics Summer School (Ven, Denmark)	2025
COLFUTURO Scholarship Grant (Top professionals of Colombia)	2019 – 2021

REFERENCES

Prof. Monika Ritsch-Marte
Medical University of Innsbruck
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Prof. Jürgen Popp
Leibniz Institute of Photonic Technology
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